OL	T	H	W	P
outlook	Temperature	Humidity	wirdy	Play
Sunny	Hot	High	Weak	No
Sunny	Mild	High	Strong	No
Cloudy	Hot	High	weak	yes
Rainy	Mild	High	Weak	yes
Rainy	Cool	Normal	weak	yes

$$\Rightarrow -\frac{4}{8}\log_2\frac{4}{8} - \frac{4}{8}\log_2\frac{4}{8}$$

$$\Rightarrow -0.5[\log_2(0.5) + 0.5\log_2(0.5)] \quad 0.25 \quad 0.46$$

$$\Rightarrow -0.5(-1) - 0.5(-1)$$

$$\Rightarrow \text{Entropy} = 1$$

$\Rightarrow$

Information gain:

Sunny: yes: 0, No = 2, Total = 2

$$-\frac{0}{3}\log_2\frac{0}{3} - \frac{2}{3}\log_2\left(\frac{2}{3}\right)$$

$$0 - 0.66\log_2(0.66)$$

$$-0.599$$

Entropy = 0

cloudy = entropy

Rainy = ent

weighted en

$\frac{2}{5} \times 0 +$

entrophy =

Informati

weak =

Weak = ye

$-\frac{3}{4}\log_2$

$= -0.75$

$\Rightarrow -0.3$

$\Rightarrow -0.$

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Strong = yes = 0, No = 1

$$\frac{0}{1}\log_2\frac{0}{1} - \frac{1}{1}\log_2\frac{1}{1}$$

Entropy = 0

$$\frac{4}{5}\log(0.189) + \frac{1}{5}(0)$$

$$0.8 \times 0.189 + 0$$

$$= 0.1512$$

Info gain of wind:

$$= 0.976 - 0.1512$$

$$= 0.5648$$

Day

Cloudy = entropy = 0

Rainy = entropy = 0

weighted entropy of weather

$\frac{2}{5} \times 0 + \frac{1}{5} \times 0 + \frac{2}{5} \times 0 = 0$

0, 716 - 01

entrophy = 0.716

Information gain of wind

weak = 4, strong = 1

weak = yes = 3, NO = 1 - Total = 4

$\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4}$

$0.75 \times 0.415 - 0.25 \log_2 (0.25)$

$0.311 - 0.25 \times -2$

$0.311 + 0.5$

$= 0.189$

Teacher's Sign.

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Lareb